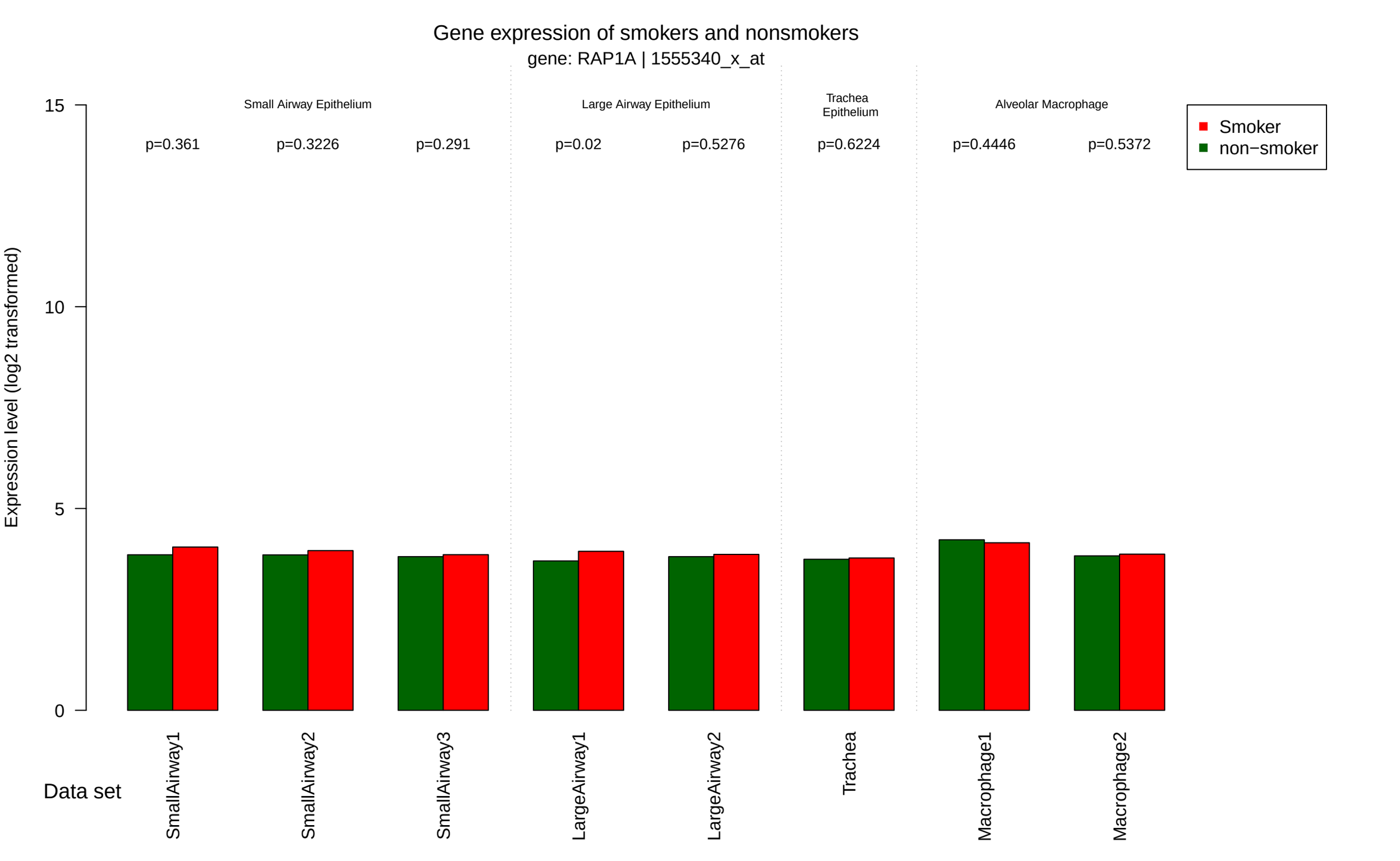
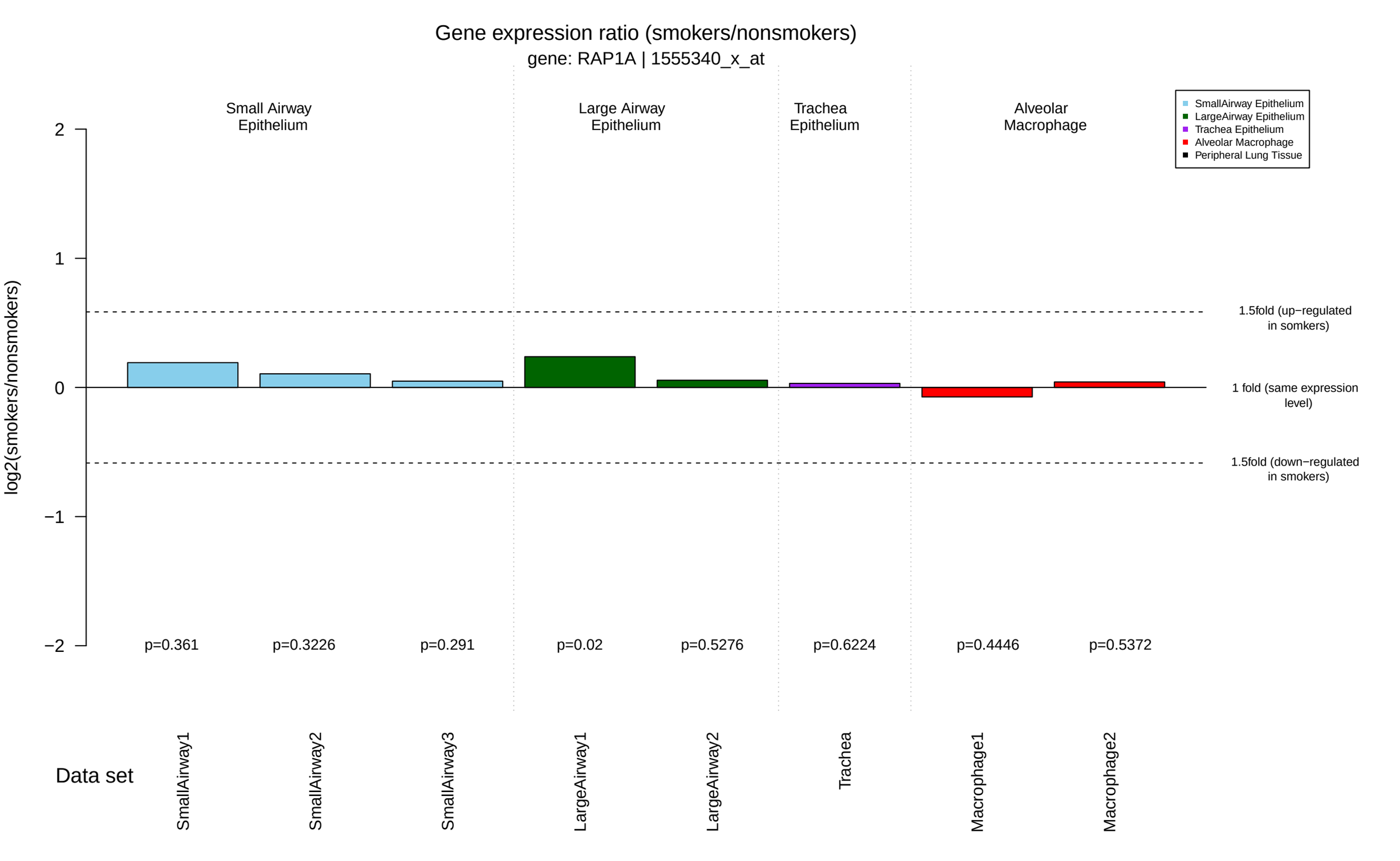
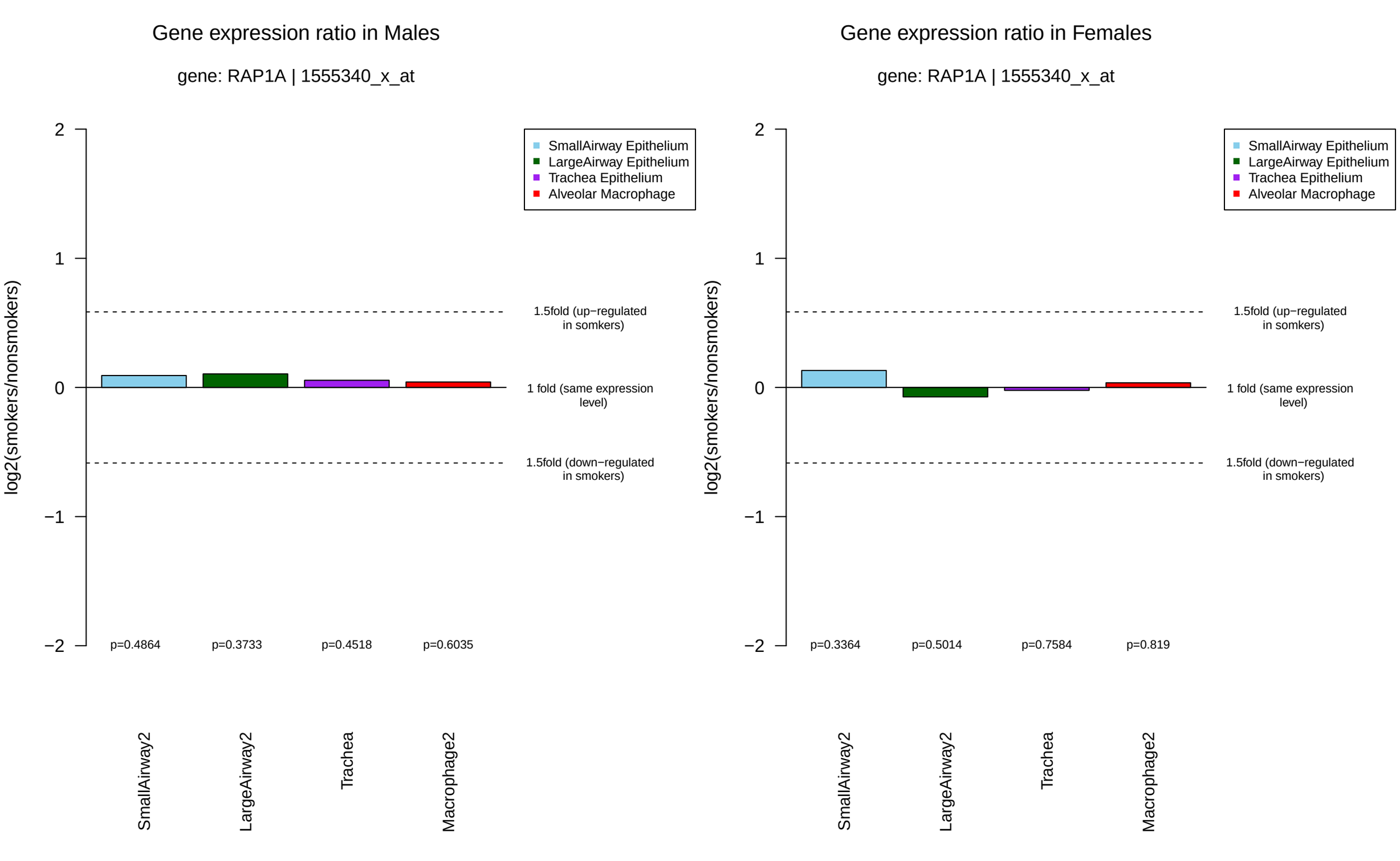
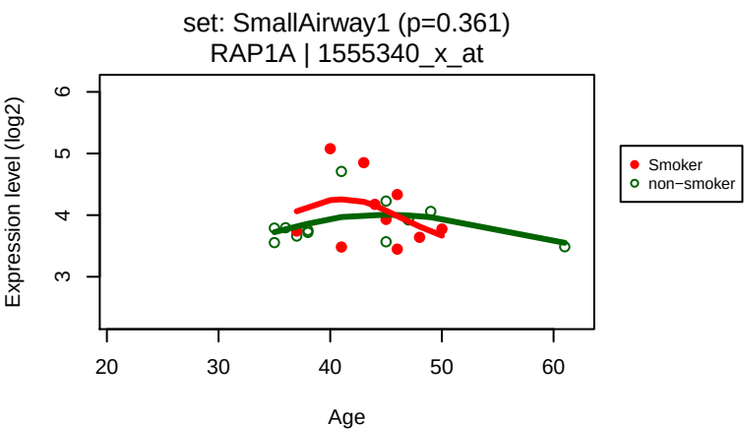
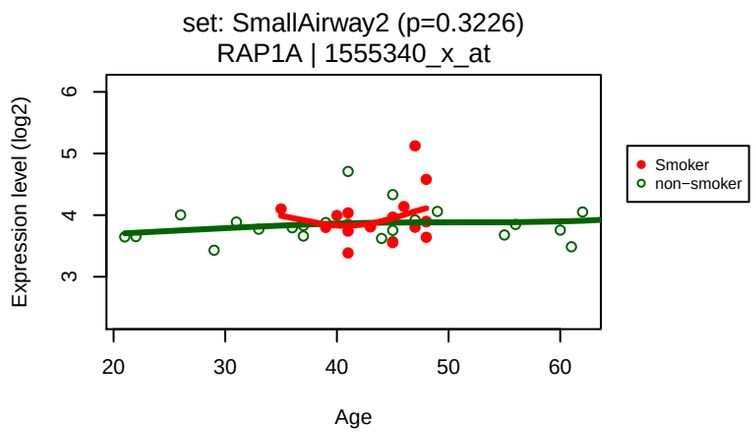
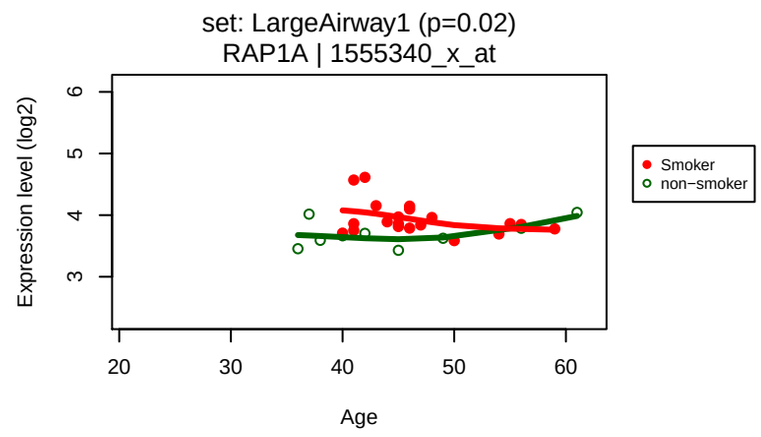

Gene expression of smokers and nonsmokers
gene: RAP1A | 1555340_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.361
p=0.3226
p=0.291
p=0.02
p=0.5276
p=0.6224
p=0.4446
p=0.5372
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

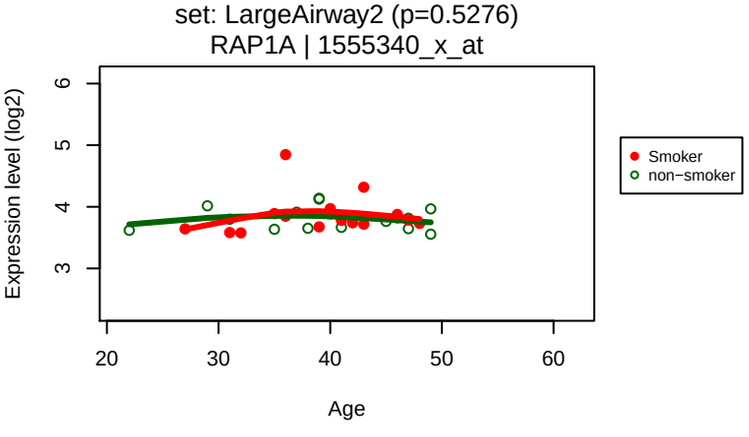

Gene expression ratio (smokers/nonsmokers)
gene: RAP1A | 1555340_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.361
p=0.3226
p=0.291
p=0.02
p=0.5276
p=0.6224
p=0.4446
p=0.5372
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

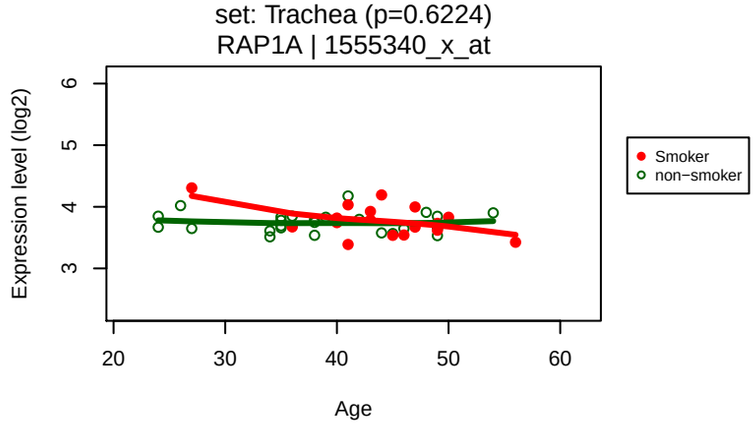

Gene expression ratio in Males
gene: RAP1A | 1555340_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4864
p=0.3733
p=0.4518
p=0.6035
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAP1A | 1555340_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3364
p=0.5014
p=0.7584
p=0.819
SmallAirway2
LargeAirway2
Trachea
Macrophage2

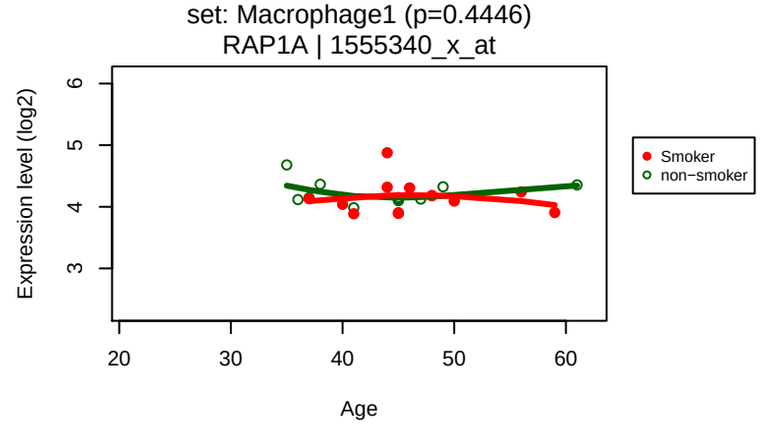

set: SmallAirway1 (p=0.361)
RAP1A | 1555340_x_at
Expression level (log2)
Age
Smoker
non−smoker

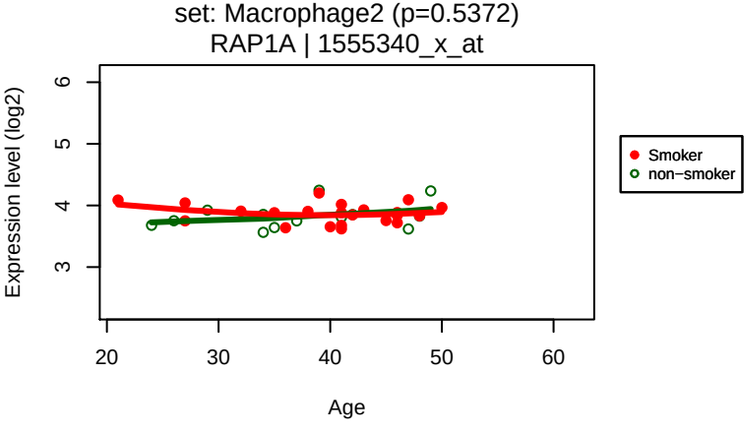

set: SmallAirway2 (p=0.3226)
RAP1A | 1555340_x_at
Expression level (log2)
Age
Smoker
non−smoker

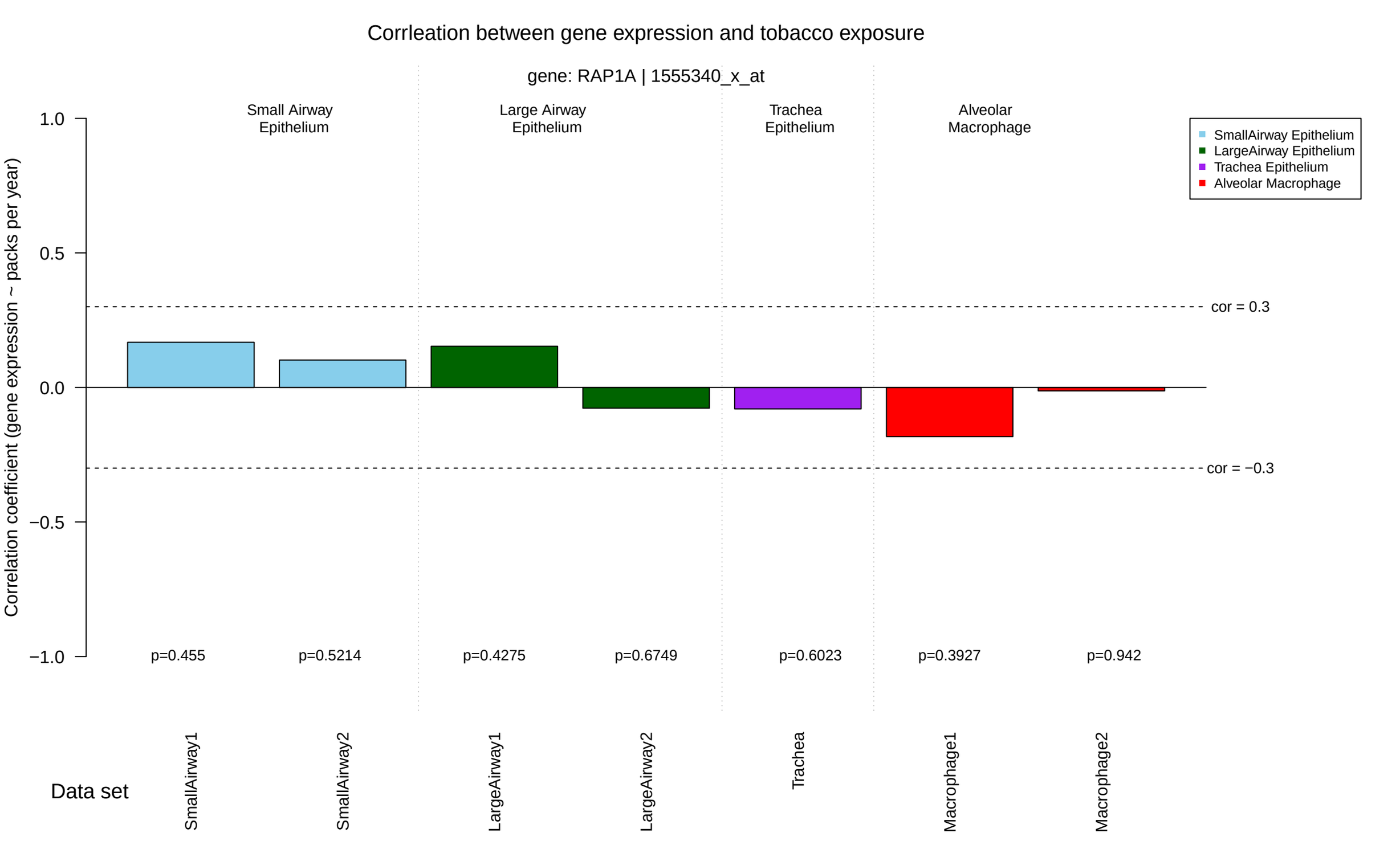

set: LargeAirway1 (p=0.02)
RAP1A | 1555340_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5276)
RAP1A | 1555340_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6224)
RAP1A | 1555340_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4446)
RAP1A | 1555340_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5372)
RAP1A | 1555340_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAP1A | 1555340_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.455
p=0.5214
p=0.4275
p=0.6749
p=0.6023
p=0.3927
p=0.942
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2